

Lucio L. Garcia, PhD

Data Scientist | Artificial Intelligence & Machine Learning | Graph Analytics & NLP

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PROFESSIONAL PROFILE

Data Scientist and PhD in Physics specialized in graph-based modeling and complex systems dynamics. Expert in processing massive unstructured data, Artificial Intelligence, advanced statistics, and Natural Language Processing (NLP). Proven track record in quantifying information flows and collective behavior, translating scientific research into strategic insights and scalable technological solutions.

PROFESSIONAL EXPERIENCE

CONICET – University of Buenos Aires

2021 – 2026

Data Scientist / Doctoral Researcher

Buenos Aires, Argentina

- **Big Data Processing:** Processed and cleaned millions of temporal interaction records using the **Python ecosystem** (pandas, NumPy, SciPy).
- **Network Science:** Designed large-scale graph models with **NetworkX** and **igraph** to detect influence patterns and community structures in social systems.
- **Machine Learning & NLP:** Developed AI and text processing pipelines to identify ideological clusters and agenda-setting dynamics using probabilistic models.
- **Business/Research Metrics:** Created topological anomaly detection algorithms for real-time monitoring of public opinion.
- **Selected Publications:** Spatiotemporal modeling of information diffusion (*Chaos, AIP*) and behavioral clustering in networks (*EPJ Data Science*). [\[Google Scholar Profile\]](#)

Federal University of Viçosa (UFV)

2025

Visiting Researcher (CAPES-MOVE Fellow)

Minas Gerais, Brazil

- Implemented comparative information diffusion models between media markets in Argentina and Brazil.
- Optimized simulation algorithms for complex networks, improving computational efficiency in news propagation analysis.

University of Buenos Aires / Torcuato Di Tella University

2019 – 2023

Teaching Assistant (Data Analysis)

Buenos Aires, Argentina

- Taught computational tools and applied statistics to undergraduate and graduate students.

TECHNICAL SKILLS

- **Languages:** Python (Advanced), SQL, R, LaTeX.
- **Data & AI Stack:** pandas, NumPy, SciPy, Scikit-learn, Matplotlib, Seaborn.
- **Graph Analysis:** NetworkX, igraph, Centrality, Community Detection, Hypergraphs.
- **Specialties:** Artificial Intelligence, Machine Learning, NLP (Topic Modeling), Bayesian Statistics, Time Series.
- **Tools:** Git, Docker, Linux/Bash, Jupyter, VS Code.

EDUCATION

PhD in Physics

2021 – 2026

University of Buenos Aires (UBA) | Thesis defended with honors in April 2026.

MSc in Physics

2012 – 2020

University of Buenos Aires (UBA).

LANGUAGES

Spanish (Native), **English** (Fluent), **Portuguese** (Professional working proficiency).